



PHYSICS

Stage 2

WACE Examination 2010

Final Marking Key

This 'stand alone' version of the WACE Examination 2010 Final Marking Key is provided on an interim basis.

The Standards Guide for this examination will include the examination questions, marking key, question statistics and annotated candidate responses. When the Standards Guide is published, this document will be removed from the website.

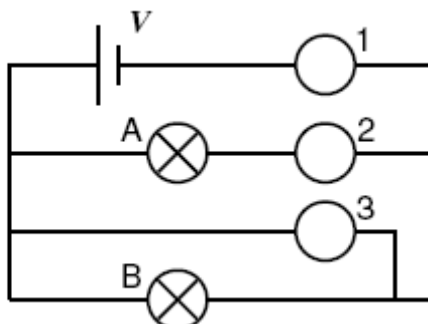
Section One: Short answers

40% (64 Marks)

Question 1

(3 marks)

A lighting circuit diagram includes a cell and two globes A and B as shown below.



- (a) Indicate in circle 1 the direction of the conventional current through this point. (1 mark)
- (b) In circles 2 and 3, place a V or A to indicate if a voltmeter or ammeter would be most appropriate to complete the circuit diagram. (2 marks)

Description	Marks
(a) ← (On the line is acceptable)	1
(b) 2 – A	1
3 – V	1
	Total 3

Question 2

(3 marks)

There are at least 37 isotopes of gold, but only one is stable. The stable isotope of gold is written as $^{197}_{79}\text{Au}$.

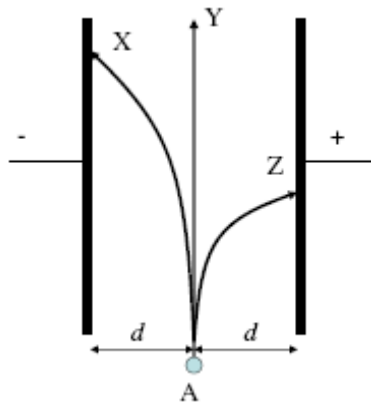
- (a) Determine how many neutrons the stable isotope of gold contains. (1 mark)
- (b) Compare what occurs to an isotope that is not 'stable' with an isotope that is 'stable'. (2 marks)

Description	Marks
(a) $197 - 79 = 118$ neutrons	1
(b) Unstable isotopes will undergo decay (give off a form of ionizing radiation) until the isotope becomes stable	1
Stable means the isotope will not undergo transmutation and form another isotope. (Must state both cases.)	1
	Total 3

Question 3

(1 mark)

There is a uniform electric field between two charged parallel plates as shown below. Three particles (α , β and γ) are ejected from A into the field parallel with the plates with similar velocities. Their paths (X, Y and Z) are shown on the diagram below.



The particles and their paths are best named as

- A X = γ Y = β and Z = α
- B X = α Y = β and Z = γ
- C X = β Y = α and Z = γ
- D X = α Y = γ and Z = β
- E X = γ Y = α and Z = β
- F X = β Y = γ and Z = α

Answer: _____

Description	Marks
D	1
	Total 1

Question 4

(3 marks)

Airships, blimps, scientific balloons and party balloons are usually inflated with helium. However, someone thinks that helium, ${}^4_2\text{He}$, being a product of radioactive decay, is unsafe. Explain, with reference to the ionisation of atoms, the relative safety of helium.

Description	Marks
Helium (atom/gas) is safe,	1
but He nuclei (alpha particles) are not. On Earth, helium is primarily a product of the radioactive decay of much heavier elements, which emit helium nuclei called alpha particles. (May say He is an ionising particle – acceptable.)	1
Alpha particles, not helium atoms, (slow down and capture 2 electrons, and become helium atoms) ionise atoms during collisions.	1
	Total 3

Question 5

(4 marks)

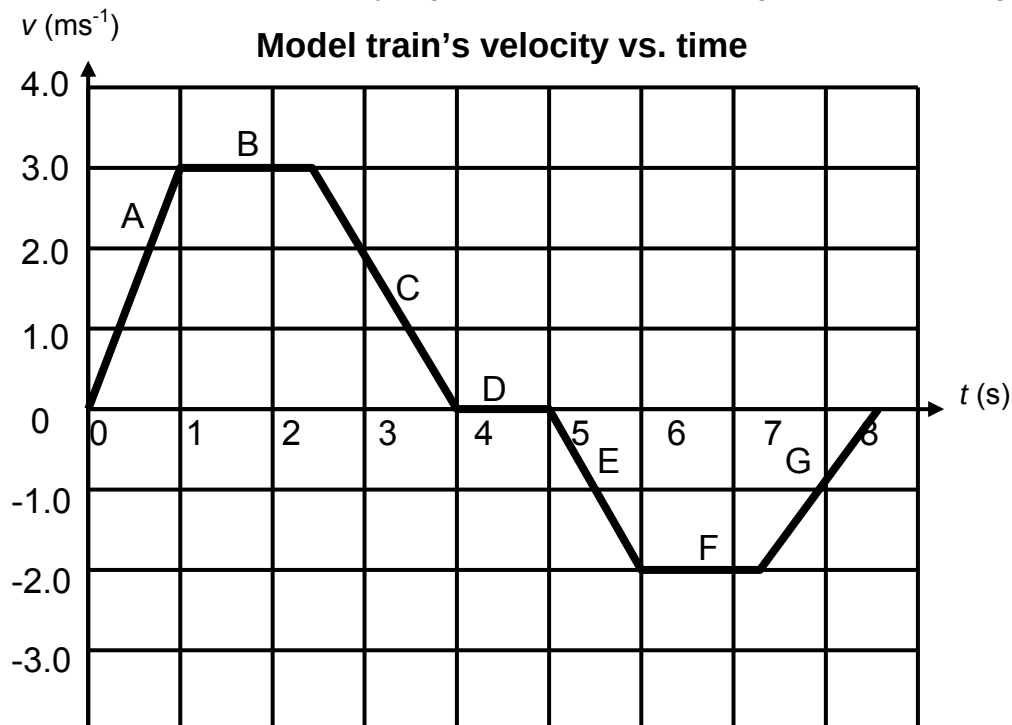
- (a) Sketch a diagram of a circuit that has a total resistance of $15.0\ \Omega$, consisting of only $10.0\ \Omega$ resistors. Include a power supply and a switch in your circuit. (2 marks)
- (b) Calculate the potential difference required to provide a total current of $1.50\ \text{A}$ through the circuit. (2 marks)

Description	Marks
(a) $R1 + (R2 \text{ in parallel with } R3)$	1
Switch and power in series (diagram)	1
(b) $V=IR=1.5 \times 15$	1
$=22.5\text{V}$	1
	Total 4

Question 6

(3 marks)

A model electric train's velocity is graphed as it moves along a section of straight track.



- (a) During which section of the graph is the acceleration the greatest? (1 mark)

Circle the correct answer: A B C D E F G

- (b) During which section is the train not moving? (1 mark)

Circle the correct answer: A B C D E F G

- (c) At the end of the journey, the train's displacement relative to its starting position will be: (1 mark)

Circle the correct answer: Positive Zero Negative

Description	Marks
(a) A	1
(b) D	1
(c) Positive	1
	Total 3

Question 7

(4 marks)

A hair dryer was used for 10.0 minutes to dry a person's wet hair. When the hair dryer was connected to a 240 V supply, it drew a current of 4.80 A.

- (a) How much charge passed through the coil of the hair dryer in this time? (2 marks)
- (b) Calculate the power of the hair dryer. (2 marks)

Description	Marks
(a) $q = It = 4.8 \times 10 \times 60$	1
$= 2.88 \times 10^3$ Coulombs	1
(b) $P = VI = 240 \times 4.8$	1
$P = 1.15$ kW	1
	Total 4

Question 8

(4 marks)

A 25.0 gram cube of ice, initially at 0.00°C, fell on a bench and melted.

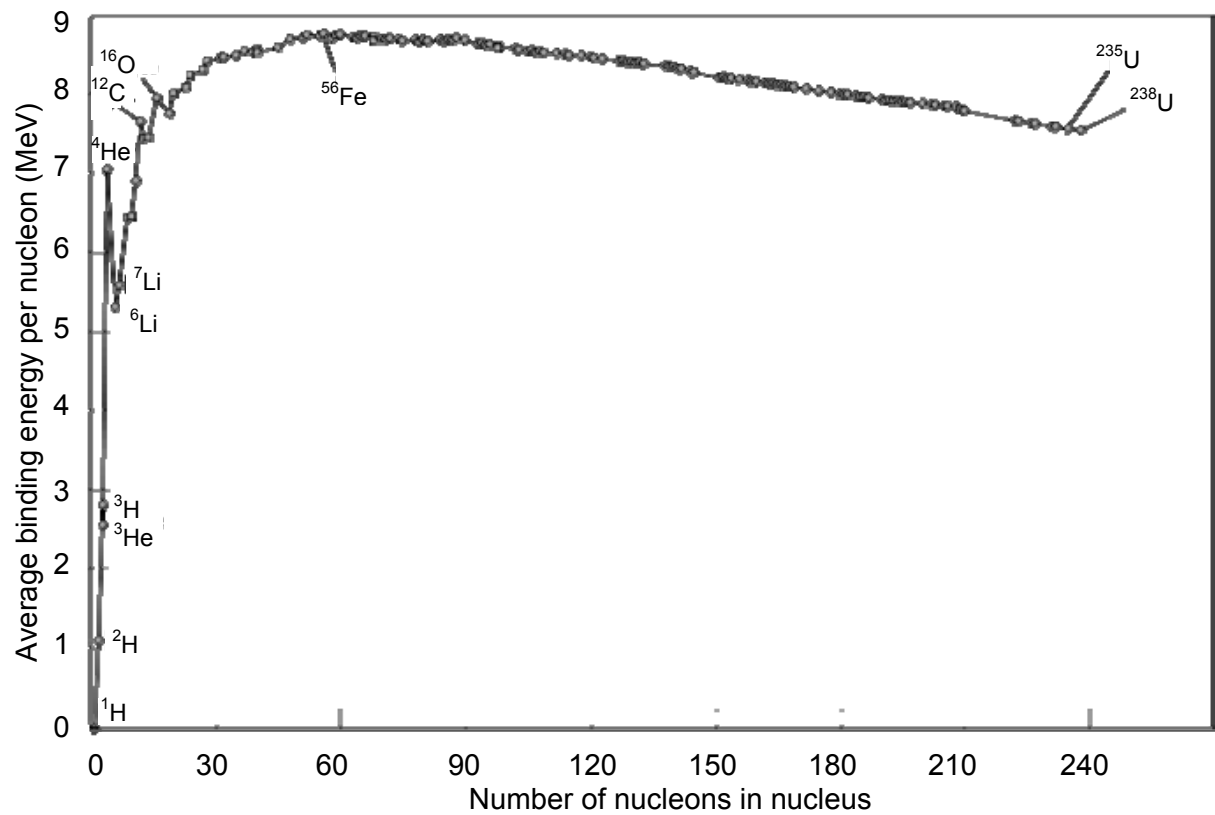
- (a) Calculate the energy required to change the ice cube from a solid to a liquid. (2 marks)
- (b) Describe briefly what happened to the particles of the bench that were in contact with the ice cube as it melted. (2 marks)

Description	Marks
(a) $Q = mL_f = .025 \times 3.34 \times 10^5$	1
$Q = 8350$ J	1
(b) The kinetic energy decreases as the energy of the bench is transferred to the ice cube. (Mentioning 'energy transfer' is acceptable.)	1
so the particles slow down.	1
	Total 4

Question 9

(3 marks)

Below is a binding energy curve for common isotopes.



(a) Which element listed is considered to be the most stable? (1 mark)

Circle the correct answer: ^3H ^4He ^{56}Fe ^{235}U

(b) Explain your choice, using information from the graph. (2 marks)

Description	Marks
(a) Fe^{56}	1
(b) Fe^{56} has the highest average binding energy per nucleon	1
The nucleons would require the most energy to break them apart making it more stable	1
Total 3	

Question 10

(3 marks)

Uranium-235, $^{235}_{92}\text{U}$, has a half life of 7.35×10^5 years and uranium-238, $^{238}_{92}\text{U}$, has a half life of 4.5×10^6 years. Compare the two isotopes of uranium, given that they both have the same number of atoms.

- (a) Which sample would have the greater weight? (1 mark)

Circle the correct answer: uranium-235 equal uranium-238

- (b) Which sample would have the greater number of decays per second? (1 mark)

Circle the correct answer: uranium-235 equal uranium-238

- (c) The isotope uranium-238, $^{238}_{92}\text{U}$, will decay into an isotope of thorium-234, $^{234}_{90}\text{Th}$. Which other particle is produced by this decay? (1 mark)

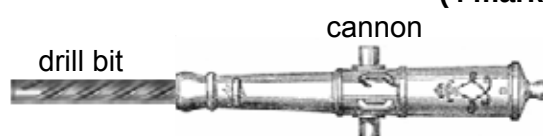
Circle the correct answer: $^4_2\alpha$ $^0_1\beta$ $^0_{-1}\beta$

Description	Marks
(a) Uranium-238	1
(b) Uranium-235	1
(c) $^4_2\alpha$	1
Total 3	

Question 11

(4 marks)

One of the original theories of heat involved the concept of heat being contained in an object and, when the object was cut, the heat contained in the object being released. Drilling holes in cannons challenged this theory. As the drill bit becomes duller, the cannon and drill bit get hotter and it takes longer to drill a hole to the right depth.



- (a) Explain why the cannon gets hotter when the drill becomes less sharp. (2 marks)

- (b) Water is used when drilling holes through rock during mining operations to keep the drill bit from overheating. If 0.150 litres of the cooling water (initially at 100°C) was evaporated, how much energy was removed from the drill? (2 marks)

Description	Marks
(a) Less cutting means more friction or more repeated collisions	1
Kinetic theory, more movement from collisions then more heat	1
(b) $Q_{\text{lost}} = mL_v = 0.150 \times 2.26 \times 10^6$	1
$= 3.39 \times 10^5$ joules	1
Total 4	

Question 12

(4 marks)

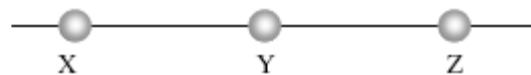
E, F, G and H are very small glass balls. F, G and H are charged, but it is not known what type of charge they possess. In order to test the types of charge on these balls, we charge glass ball E by rubbing it with silk. This removes electrons from the glass ball.

- (a) Once E has been charged, we find E attracts F and F repels G, but G attracts H. Which are two negatively charged balls? (1 mark)

- A E and F
- B F and H
- C E and G
- D F and G

Correct answer: _____

Three balls are selected from these four balls and renamed 'X', 'Y' and 'Z'. These three balls are placed in a straight line with a reasonable distance between them. Balls X and Z are fixed in place and Y is kept in equilibrium through electrostatic forces as shown below.



- (b) For Y to stay in a stable position, the charges on the spheres need to be (1 mark)

- A X is positive, Y is positive and Z is negative.
- B X is positive, Y is negative and Z is positive.
- C X is negative, Y is positive and Z is positive.
- D X is negative, Y is negative and Z is positive.

Correct answer: _____

- (c) An atom has two electrons removed from it. Which statement is correct? (2 marks)

- A The atom becomes a different isotope.
- B The atom becomes more positive.
- C The atom becomes more negative.
- D The atom is unchanged.

Correct answer: _____

Explain your choice.

Description	Marks
(a) D	1
(b) B	1
(c) B	1
By removing negative charges, there are more positives charges, making the atom more positive. (‘Electrons are lost or transferred’ is acceptable.)	1
	Total 4

Question 13

(6 marks)

Thorium, $^{228}_{90}\text{Th}$, is a radioactive isotope that undergoes decay via a series of steps to eventually reach bismuth, $^{212}_{83}\text{Bi}$, a stable isotope. Each of these steps involves the release of ionising radiation.

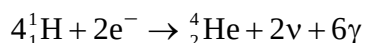
- (a) Determine how many alpha $^4_2\alpha$ and beta $^0_{-1}\beta$ particles in total are released in the transitions between these two isotopes. Show your working. (3 marks)
- (b) In an experiment to study radioactive decay, a 100 gram sample of thorium is placed in a sealed container to decay. After a suitable length of time, the thorium atoms have become mostly bismuth atoms. The container is regularly analysed over this time. Describe how the number of decays per second and the weight of the container change during this time. (3 marks)

Description	Marks
(a) $228-212=16$ mass difference means $16/4=4$ alpha particles	1
$90-83=7$ protons difference, but 8 from alpha decay,	1
so one beta decay to change neutron to proton	1
(b) The greatest number of decays per second occur at the beginning when there is a higher amount of unstable material	1
The weight will remain equal over the length of the experiment	1
The decayed particles have been kept in the container (could say matter has changed into energy, so is slightly lighter)	1
	Total 6

Question 14

(4 marks)

A major source of energy in the Universe is a product of fusion reactions. These include many steps and are very complex. One reaction involves the combination of four protons and two electrons to form a helium nucleus, two neutrinos and six photons. Neutrinos and photons have no mass. The overall equation for this reaction can be written as



- (a) Use the information listed in the table below to calculate the energy (in MeV) released from this process due to the mass defect. (2 marks)

Particle	Proton	Helium	Electron	Neutron
Mass (u)	1.007276	4.002602	0.000549	1.008665

- (b) The major component of the Sun is ^1_1H , which enables fusion reactions to take place. Explain why this is important to us on Earth. (2 marks)

Description	Marks
(a) $\Delta m = [4(1.007276) + 2(0.000549) - 4.002602] = 0.0276\text{u}$	1
$\Delta E = \Delta m c^2 (931 \text{ MeV}) = 25.696 \text{ MeV (or } 25.7 \text{ MeV)}$	1
(b) The temperature inside the sun is very high and converts the hydrogen into helium.	1
This energy is radiated out and is the main source of energy for the earth	1
	Total 4

Question 15

(5 marks)

The average power supplied by an adult's heart for circulating blood is about 1.5 watts.

- (a) How much work does an adult human heart do in one hour? (2 marks)
- (b) If this amount of work is used to lift a 50.0 kg object with a constant velocity, what is the theoretical maximum height to which the object can be raised? Use an energy value of 5000 J if you were unable to determine an answer for (a). (3 marks)

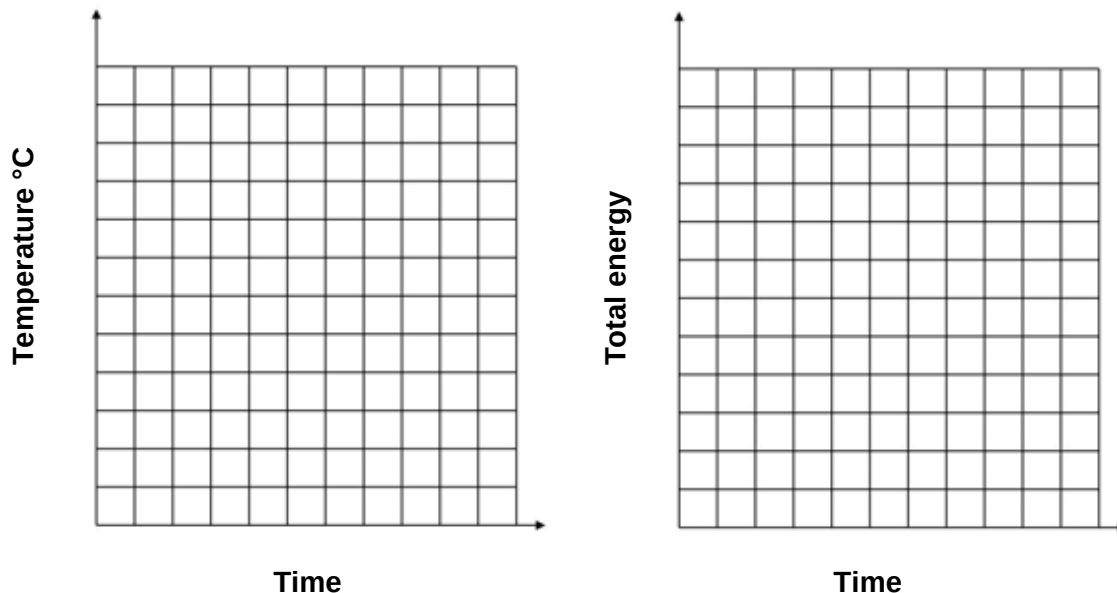
Description	Marks
(a) $W = P \times t = 1.5 \times 60 \times 60$	1
$= 5400 \text{ J}$	1
(b) $W = mgh$; $h = W / mg$	1
$= 5400 / (50 \times 9.8)$	1
$= 11 \text{ m}$	1
	Total 5

Question 16

(5 marks)

An experiment is conducted in which 500 mL of ice at 0.00°C is placed into a beaker. An electric heater with a constant power output is used to heat the beaker.

- (a) Sketch graphs on the axes below showing the temperature change over time, and the energy input over time, as the ice is melted and then brought to a boil. Label melting point and boiling point on the temperature graph. (3 marks)



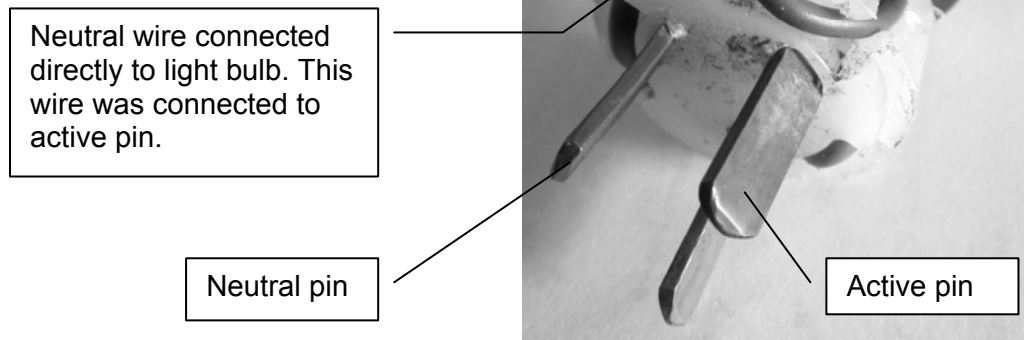
- (b) The experiment was repeated with a small fan placed on the top of the beaker. Explain how this would affect the melting and boiling results. (2 marks)

Description	Marks
(a) Temperature graph – flat, positive linear, flat	1
Any 2 points (minimum) included, e.g. melting, heating, boiling	1
Total energy input, should be positive linear	1
(b) fan would increase melting rate as warm air flow increases	1
Though boiling would take longer as fan cools water	1
	Total 5

Question 17

(5 marks)

A person complained to an electrician about getting a shock when changing a light globe in a lamp even though the lamp had been switched off. The electrician examined the plug and found the active and neutral wires had been swapped around in the plug and explained that appliance switches are only designed to operate on the active wire.



- (a) Describe the conditions that led to the person receiving the shock. (3 marks)
- (b) Name a safety feature in modern houses that limits the risk of receiving an electric shock, and describe how it works. (2 marks)

Description	Marks
(a) the switch on the neutral side of the circuit had broken the circuit, so the light was not on.	1
The switch is designed to be on the active wire and not neutral, so by swapping them at the plug the current was delivered to where the globe was. Active wire has the potential to supply current to the globe.	1
Any contact with the metal side, even though it is off, will result in current trying to ground through the person Completes a circuit – acceptable.	1
(b) Any feature, e.g. elcb, rcd, etc. (Acronyms are acceptable.)	1
Good description of the feature, eg measures current in/out and breaks circuit if not equal	1
	Total 5

End of Section One

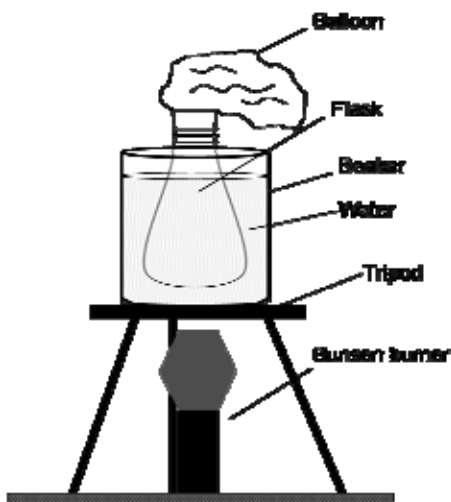
Section Two: Problem-solving

50% (80 Marks)

Question 18

(12 marks)

An empty, clean and clear flask was put into a refrigerator for one hour. The flask was then taken from the refrigerator and a balloon was placed on the open end of the flask. The flask was then placed into a beaker with hot water, as shown below.



- (a) Explain what was seen to happen to the balloon as the gas in the flask absorbed the thermal energy from the hot water. (2 marks)

Description	Marks
the volume increases so the balloon gets bigger	1
Particles move faster and spread out	1
	Total 2

- (b) Indicate the form of energy transfer (conduction, convection or radiation) that occurred in the following situations: (3 marks)

- (i) between the Bunsen burner flames and the tripod: _____
- (ii) between the tripod and the beaker: _____
- (iii) between the water and the flask: _____

Description	Marks
(i) Convection or radiation	1
(ii) Conduction	1
(iii) Conduction	1
	Total 3

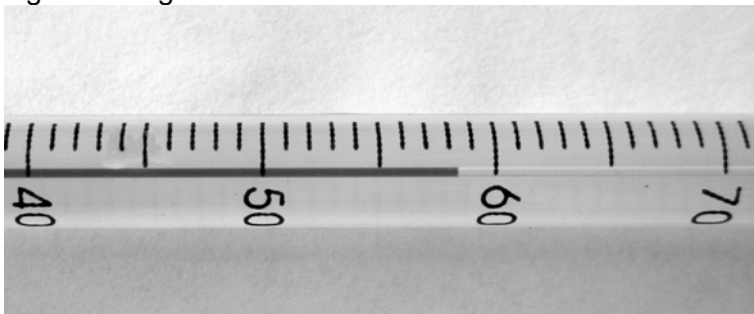
- (c) Consider the following statements describing what happens when the water and the gas in the balloon are at the same temperature. (2 marks)

- A The water has more internal energy overall than the gas.
B The average energy of the water's molecules is greater than the average energy of the gas molecules.
C The heat will flow from the water to the gas.

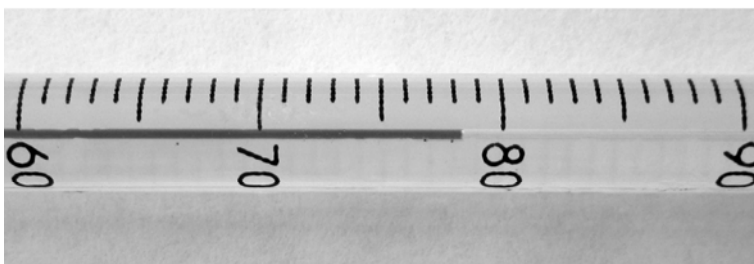
Which of the above statements is or are true? _____

Description	Marks
A (one mark off for each error)	1-2
	Total 2

- (d) A thermometer was used to measure the temperature of the water in degrees Celsius at two different times, labelled 'before' and 'after'. Determine the readings, then calculate the temperature change and the Absolute temperature uncertainty. Use appropriate significant figures. (5 marks)



Temperature before: _____



Temperature after: _____
Temperature difference _____
Absolute temperature uncertainty: _____

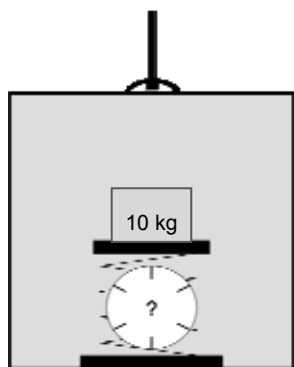
Description	Marks
58°C (May be read as 58.3 – accept 58.0 – 58.5)	1
78°C (May be read as 78.2 – accept 78.0 – 78.5)	1
20°C (May be 19.9 – 3 sig figs)	1
$\pm 0.5 + \pm 0.5 = \pm 1.0^{\circ}\text{C}$	1
Observes appropriate significant figures (depends on whether 2 or 3 sig figs used in measurements.)	1
	Total 5

Question 19

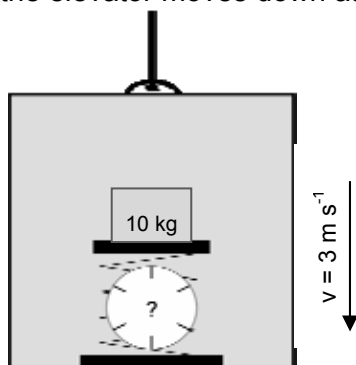
(11 marks)

An experiment is conducted in which an object with a mass of 10 kg is placed on a scale sitting on the floor of an elevator, as shown in the diagrams below. Assuming the local gravity is 9.8 m s^{-2} , estimate the reading of the scale in newtons when

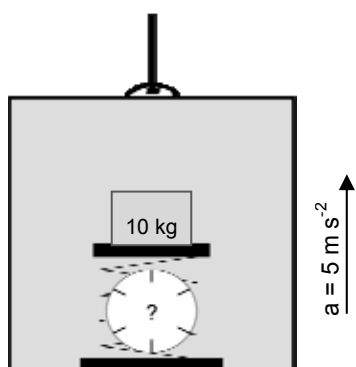
- (a) (i) the elevator is stationary; (1 mark)



- (ii) the elevator moves down at a constant velocity; (2 marks)



- (iii) the elevator accelerates up at a constant acceleration. (2 marks)

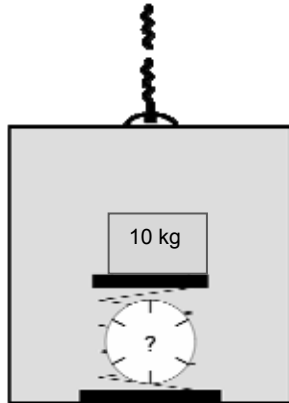


Description	Marks
(i) $W = mg = 10 \times 9.8 = 98 \text{ N}$	1
(ii) Recognise $a = 0$	1
$= 98 \text{ N}$	1
(iii) $W = m(g-a) = 10 \times (9.8 + 5)$	1
$= 148 \text{ N}$	1
	Total 5

- (b) During the experiment, while the elevator was accelerating up at a constant acceleration, the cable suddenly broke and the elevator began to fall freely.

- (i) Determine the new scale reading.

(2 marks)



- (ii) The elevator was moving upward at 5.30 m s^{-1} when the cable broke. It took 1.50 s to reach the ground. How high above the ground was the elevator when the cable broke? (2 marks)

Description	Marks
(b) (i) $W_{\text{after}} = 10 \times 0$	1
$= 0 \text{ N}$ (Goes from 148 N to zero – no reaction force between the scales and the lift – explanation not necessary.)	1
(ii) $s = ut + \frac{1}{2}at^2 = 5.3 \times 1.5 + \frac{1}{2}(-9.8) \times (1.5)^2$	1
$= -3.075 \text{ m}$	1
Total 4	

- (c) Modern elevators are equipped with speed governors that apply a braking friction when a maximum speed is exceeded. These brakes bring the elevator slowly to a stop. Explain why it is desirable for the elevator to be brought to a stop slowly rather than quickly. (2 marks)

Description	Marks
The change in momentum is constant, $Ft = m(v-u) = \text{constant}$ May use $F = \Delta p/t$	1
Increasing the time (slowly) decreases the force on the occupants	1
Total 2	

Question 20

(15 marks)

A set of 16 party lights is purchased to decorate the back patio of a house for a birthday party. When all lights are functional they draw a current of 3.20 A from the 24.0 V transformer supplied. When one globe is removed, half of the globes go out, leaving the other half working. When one of these working globes is removed, the remaining seven working globes go out.

- (a) Explain why the other seven globes went out when the second globe was removed, but not when the first globe was removed. (2 marks)

Description	Marks
The two lines are in parallel so only one side goes out	1
The remaining globes are in series, remove one and stop the current flow	1
	Total 2

- (b) Draw a simple circuit diagram to show how to wire all 16 globes to the 24 V power supply. (2 marks)

Description	Marks
2 parallel lines of	1
8 globes in series	1
	Total 2

- (c) Determine the voltage across each globe. (2 marks)

Description	Marks
$V_T = 8 \times V$; $V = V_T/8 = 24/8$	1
$V = 3.00V$	1
	Total 2

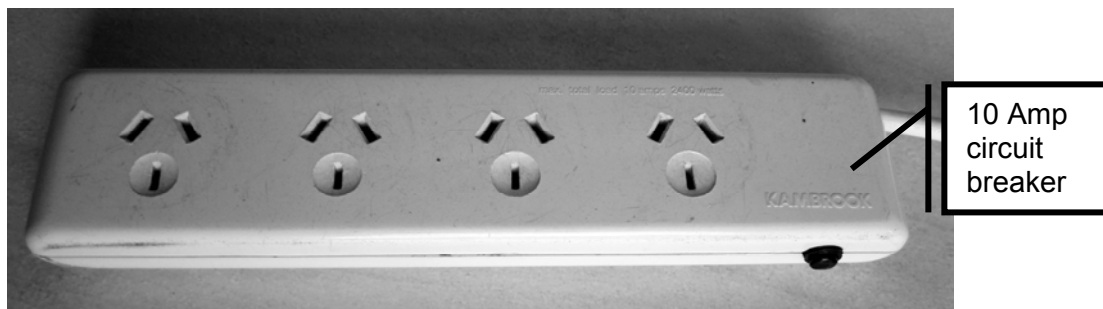
- (d) Determine the current through each globe. (2 marks)

Description	Marks
I through each strand is $\frac{1}{2}$ of 3.20 A	1
$I = 1.60A$	1
	Total 2

- (e) Calculate the power consumed by each globe. (2 marks)

Description	Marks
$P = VI$	1
$P = 3.00 \times 1.60 = 4.80 W$	1
	Total 2

- (f) If you wanted to have more than one set of lights, you might use a power board similar to the one below. This power board has a 10.0 A circuit breaker built into it, as shown in the picture.



- (i) How many sets of these party lights can operate from the power board before the circuit breaker is overloaded? (2 marks)
- (ii) Explain the purpose of the circuit breaker. (2 marks)
- (iii) Is the circuit breaker connected to the power circuit in series or in parallel? (1 mark)

Description	Marks
(i) Current drawn must be less than 10 A	1
So 3 sets of party lights would draw $3 \times 3.2 = 9.6\text{A}$	1
(ii) protects from overloading /over heating	1
By breaking the circuit when too much current is drawn	1
(iii) Must be in series with circuit	1
	Total 5

Question 21

(13 marks)

When a uranium-235 nucleus absorbs a neutron, many fission products are possible. One such reaction in a nuclear power plant results in the formation of lanthanum-148 (La), bromine-85 (Br) and neutrons.

- (a) Write an equation for this reaction and identify clearly the number of neutrons produced. (2 marks)

Description	Marks
${}^1_0\text{n} + {}^{235}_{92}\text{U} \rightarrow {}^{148}_{57}\text{La} + {}^{85}_{35}\text{Br} + 3{}_^1_0\text{n} + \text{energy}$	1–2
	Total 2

- (b) How do the neutrons released in this reaction differ from those that took part in the initial fission reaction? (1 mark)

Description	Marks
The released neutrons are much more energetic (have more speed).	1
	Total 1

Many of the products of such fission reactions are themselves radioactive but are not able to be used as an energy source for the reactor. This waste is taken from the site and stored permanently in a safe and secure place where its activity can be monitored. The measured activity from some radioactive waste when it was first removed from the reactor was 128 Bq above the background count of 2.00 Bq.

- (c) Explain what is meant by the term 'background count' and give an example of a source that contributes to it. (2 marks)

Description	Marks
There are naturally occurring sources of radiation in the environment.	1
e.g. cosmic rays, rocks such as granite, atomic testing, nuclear sources, etc	1
	Total 2

- (d) If the average half-life of the waste in part (c) is taken as being 7.00×10^5 years, calculate how long it will take for its activity to reach the same level as the background count. (4 marks)

Description	Marks
$A = A_0(1/2)^n$; $2 = 128(1/2)^n$	1
$n = 6.0$ half-lives	1
Age = $n \times t_{1/2} = 6.0 \times 7 \times 10^5$ yrs	1
$= 4.2 \times 10^6$ years	1
	Total 4

The safety device worn by an 85.0 kg nuclear power plant worker indicated that they absorbed 24.0 J of energy overall when exposed to this waste fuel during one work period.

- (e) Calculate the dose they absorbed. (2 marks)

Description	Marks
Absorbed Dose = $E/m = 24 / 85$ $\frac{t}{m} = \frac{24}{85}$	1
$= 0.282$ Gy	1
	Total 2

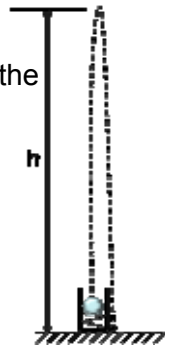
- (f) Determine the dose equivalent for the worker, assuming all of the absorbed radiation is from gamma rays. (2 marks)

Description	Marks
Dose Equivalent = Absorbed Dose $\times$ Quality Factor = 0.282×1	1
$= 0.282$ Sv	1
	Total 2

Question 22

(16 marks)

A vertical cylinder contains a spring. A ball sits on the top of the spring, as shown in the diagram. The length of the cylinder is 0.500 m and the uncompressed length of the spring is 0.400 m. The spring is then squashed down to 0.100 m and released. The kinetic energy of the ball ($m = 0.500$ kg) when it leaves the spring is 100 J.



- (a) Calculate the average force that the spring exerts on the ball to launch it. (3 marks)

Description	Marks
(a) $W = Fs \Rightarrow F = \frac{W}{s}$	1
$= \frac{100}{0.300}$	1
$F = 333\text{N}$ upward	1
Total 3	

- (b) Calculate the velocity of the object when it leaves the top of the spring. (2 marks)

Description	Marks
$E_k = \frac{1}{2}mv_0^2$; $100\text{J} = \frac{1}{2} \times 0.5 \times v_0^2$;	1
$v_0 = 20\text{ m s}^{-1}$ upward	1
Total 2	

- (c) Determine the height, h , the object reaches above the ground. (4 marks)

Description	Marks
$v^2 = u_0^2 + 2as$; or $E_k = E_p = m \times g \times s$;	1
$s = 20^2 / (2 \times 9.8)$ or $s = \frac{100}{4.9}$	1
$s = 20.4\text{m}$	1
$h = s + 0.400 = 20.8\text{ m}$ above the ground	1
Total 4	

- (d) Determine the acceleration of the object at the highest point. (1 mark)

Description	Marks
$a = g = -9.8\text{ m s}^{-2}$ down	1
Total 1	

- (e) Determine the final velocity of the object, the instant before it hits the ground. (3 marks)

Description	Marks
$v_f^2 = u_0^2 + 2ah$ $v^2 = u^2 + 2as$	1
$v_f^2 = 0 + 2(9.8) \times (20.4 + 0.40)$;	1
$v_f = -20.2\text{ m s}^{-1}$ down (-1 if no direction)	1
Total 3	

- (f) Calculate the time needed for the object to reach the ground from the highest point. (3 marks)

Description	Marks
$v_f = u_0 + at$;	1
$t_2 = -20.2 / -9.8$	1
$= 2.06\text{ s}$	1
Total 3	

Question 23

(13 marks)

A modified Joule's heat apparatus, shown in the figure below, can be used to determine the specific heat capacity of an unknown liquid. A 40.0 watt 12.0 volt DC motor provides the necessary energy input to the oil through the motion of the cable and paddle. A student wants to use this apparatus to distinguish whether a liquid is olive oil or generic vegetable oil. 0.500 kg of oil is used in the experiment.

- (a) Draw a free body diagram showing the forces acting on the frictionless pulley. (2 marks)

Description	Marks
Horizontal and Vertical Tensions	1
Force from 45° beam	1
	Total 2

- (b) During one trial, the DC motor winds up the cable at a constant velocity for 50 seconds. Calculate the work done by the motor. (2 marks)

Description	Marks
$W_{\text{DC motor}} = Pt = 40 \times 50$	1
$= 2000 \text{ J}$	1
	Total 2

- (c) Using previous experimental results, the device was determined to have an 80% energy conversion efficiency from the electric motor to the oil.

- (i) Explain what 'efficiency' means in this context. (2 marks)

- (ii) Using information from parts (b) and (c), calculate the energy input into the oil. (1 mark)

- (iii) What is one possible source of energy loss in the system? (1 mark)

Description	Marks
(i) That some of the energy is lost to the environment	1
Only 80% of the energy makes it from the motor to the oil	1
(ii) $2000 \times 0.8 = 1600 \text{ J}$	1
(iii) any reasonable answer (Friction at the spindle; some heat lost to the environment.)	
	Total 4

- (d) After several more trials, the results of this experiment are averaged and can be summarised as

'the temperature of this oil is raised 1.7°C from an energy input to the oil of $1.8 \times 10^3 \text{ J}$ '.

- (i) Calculate the specific heat capacity of the liquid using these values. (3 marks)

- (ii) A search on the internet finds the specific heat capacities for olive oil and generic vegetable oil listed as 1.97×10^3 and $1.67 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$, respectively. Explain which oil you think is in the apparatus. (2 marks)

Description	Marks
(i) $Q = mc\Delta T \Rightarrow c = Q / m\Delta T$	1
$= 1800 \text{ J} / 0.5 \text{ kg} \times 1.7^\circ\text{C}$	1
$= 2.12 \text{ (kJ/kg K)}$	1
(ii) Olive oil. It is closer to theoretical value.	1-2
	Total 5

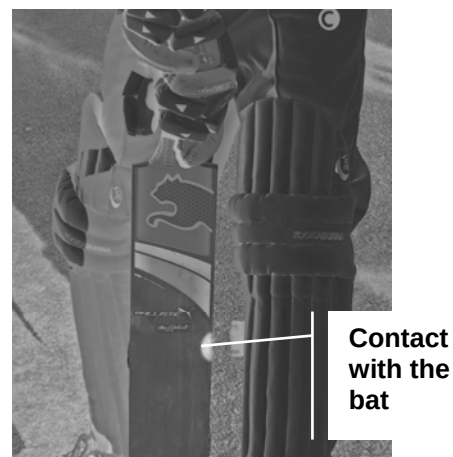
Section Three: Comprehension

10% (16 Marks)

Question 24

In cricket, a batsman is judged 'out' if the ball is caught after hitting the bat. The batsman is 'not out' if the ball is caught after hitting the batsman's leg. A cricket umpire must sometimes decide whether the ball hit the bat, the batsman's leg, or if both have been hit, which happened first. This can be tricky because the bat and the leg may be very close together, and the contact occurs over a very short time.

A technology called Hot Spot can be used to resolve this issue. Hot Spot is an infra-red imaging system, and is used to determine where, or what, the ball actually hit. There are two Hot Spot infra-red cameras, one at each end of the cricket ground. These measure and record the temperature of the bat and the batsman, before and after the ball makes contact. The infra-red images are then processed by a computer to show temperature differences between the 'before' and 'after' images.



The point is to show accurately whether the ball has hit the bat, the batsman's glove, the batsman's leg, or none of these. The black-and-white images produced by Hot Spot can potentially allow an umpire to precisely localise the ball's point of impact, and so reduce the risk of making an incorrect decision.

- (a) Using the image above as a reference, which would have the higher temperature? (1 mark)

Circle the correct answer: a light part a dark part not enough information

Description	Marks
(a) The light part has the most energy and highest temperature	1
	Total 1

- (b) Explain how the infra-red cameras are able to 'sense' where the contact or collision point has occurred. (2 marks)

Description	Marks
More infrared radiation comes out of the contact point than from the surroundings. Friction comes from collision → heat	1
The radiation travels to the sensor in the camera. Camera detects a change in temperature between the before and after image.	1
	Total 2

- (c) The increase in heat energy of the contact point only lasts for a short time. Explain one form of heat transfer that is likely to occur in this situation. (3 marks)

Description	Marks
The spot is hotter than the surrounding environment and radiates this energy back into the cooler environment. Conduction to the rest of the bat + convection into the atmosphere also acceptable.	1
Radiation – radiates more energy than takes in from environment; Conduction – passes kinetic energy into the air or bat molecules through collisions; or Convection – heats the air making it less dense and rise to be replaced with cooler air	1-2
	Total 3

- (d) A 161 gram cricket ball moving at 25.0 m s^{-1} hits the edge of a stationary bat transferring 18.1 J of energy to the bat.

- (i) Calculate the speed of the ball after the collision. (4 marks)
- (ii) At the moment the photograph is taken, about 4 grams of wood absorb the 18.1 J of energy and increase in temperature. Estimate the temperature increase of this portion of the bat, given that the specific heat of the bat is $2.25 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$. Give your answer to an appropriate number of significant figures. (4 marks)

Description	Marks
(i) $\Delta E_k = \frac{1}{2} mv^2 - \frac{1}{2} mu^2$	1
$\frac{1}{2} mv^2 = \Delta E_k - \frac{1}{2} mu^2$	1
$v = \sqrt{[(\Delta E_k - \frac{1}{2} mu^2)/(\frac{1}{2} m)]}$	
$v = \sqrt{[(18.1 - \frac{1}{2} \times 0.161 \times 25^2)/(\frac{1}{2} \times 0.161)]}$	1
$v = 20.0 \text{ m s}^{-1}$	1
(ii) $Q = mc(T_f - T_i)$	1
$(T_f - T_i) = Q / mc$	1
$(T_f - T_i) = 18.1 / (0.004 \times 2250)$	
$= 2.011111^\circ\text{C}$	1
$= \text{about } 2^\circ\text{C given limiting sig figs}$	1
	Total 8

- (e) After the ball is struck, it rolls along the ground and comes to a stop. Use Newton's Second Law to explain this. (2 marks)

Description	Marks
Newton's Second Law, $F = ma$, explains that an unbalanced force will result in an object changing its motion (acceleration)	1
The unbalanced force comes from the friction between the ball and the grass/ground, slowing the ball.	1
	Total 2

End of questions

ACKNOWLEDGEMENTS

Section One:

Question 9 Binding energy curve graph from: Fastfission. (2009). Retrieved November 18, 2009 from
http://commons.wikimedia.org/wiki/File:Binding_energy_curve_-_common_isotopes.svg.

Question 11 Image from: Encyclopedie Larousse Illustree. (1897). Canon_de_16_Gribeauval_pour_les_sieges. Retrieved April 29, 2010 from
http://en.wikipedia.org/wiki/File:Canon_de_16_Gribeauval_pour_les_sieges_before_1923.jpg#file.

Section Three:

Question 24 Adapted from: Hot Spot (cricket). (2010, February 5). *Wikipedia*. Retrieved February 22, 2010, from
[http://en.wikipedia.org/w/index.php?title=Hot_Spot_\(cricket\)&oldid=342080831](http://en.wikipedia.org/w/index.php?title=Hot_Spot_(cricket)&oldid=342080831).